The Calabi–Yau-to-chiral functor and the $\kappa_\bullet$-spectrum

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Abstract

Where do chiral algebras come from? Volumes I and II of the modular Koszul duality programme accept a chiral algebra as given and extract its invariants: bar-cobar adjunction, the modular characteristic κ_{ch}, the shadow tower, the five theorems A–D+H. The geometric source is left unspecified. This paper constructs it.

A Calabi–Yau category $\mathcal{C}$ of dimension d carries a cyclic A_∞-structure: a non-degenerate trace $\mathrm{Tr}\colon \mathrm{HC}^-_d(\mathcal{C}) \to k$ on negative cyclic homology. We construct a functor

$$\Phi\colon \mathrm{CY}_d\text{-Cat} \longrightarrow E_2\text{-ChirAlg}$$

in four steps: extract the Lie conformal algebra $\mathfrak{L}_{\mathcal{C}}$ from the Gerstenhaber bracket on $\mathrm{HH}^\bullet(\mathcal{C})$; pass to the factorization envelope $\mathrm{Fact}_X(\mathfrak{L}_{\mathcal{C}})$ on a curve X; enhance via the $\mathbb{S}^d$-framing to an E_n-chiral algebra; quantize using the CY trace. The functor is proved at $d = 2$ (Theorem CY-A$_2$) and at $d = 3$ (Theorem CY-A$_3$, ∞-categorical proof).

A single CY manifold admits multiple chiral algebraizations, each with its own modular characteristic. We introduce the $\kappa_\bullet$-spectrum $\mathrm{Spec}_{\kappa_\bullet}(X) = \{\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}\}$: four numbers measuring four faces of the same manifold. For $K3 \times E$: $\mathrm{Spec}_{\kappa_\bullet} = \{2, 3, 5, 24\}$. The shadow–Siegel gap theorem identifies four structural obstructions separating $\kappa_{\mathrm{ch}} = 3$ from $\kappa_{\mathrm{BKM}} = 5$.

For toric CY$_3$, the functor interfaces with the critical cohomological Hall algebra: $\mathrm{CoHA}(Q_X, W_X) \simeq Y^+(\widehat{\mathfrak{gl}}_{Q_X})$ provides the E_1-sector, and the Drinfeld center recovers E_2-braiding. The E_1 stabilization theorem proves that for $d \geq 3$, the CY chiral algebra is natively E_1: the Schouten–Nijenhuis bracket vanishes, and all noncommutative structure arises from quantization.

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1 Introduction

1.1 The source problem

A chiral algebra is a solution. What is the problem it solves?

The bar-cobar machine of Volume I extracts from any chiral algebra A on a curve X a sequence of invariants: the bar complex $B(A) = T^c(s^{-1}\bar{A})$ with its deconcatenation coproduct, the universal Maurer–Cartan element Θ_A satisfying $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, the modular characteristic $\kappa_{\mathrm{ch}}(A)$ governing the genus-1 curvature, and the shadow tower $\{S_r\}_{r\geq 2}$ encoding higher-genus data. The five theorems (A–D+H) are structural properties of this machine. Volume II interprets the output physically: the derived chiral center $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = C^\bullet_{\mathrm{ch}}(A, A)$ is the bulk of a 3d holomorphic-topological gauge theory; the $\mathrm{SC}^{\mathrm{ch,top}}$ pair $(C^\bullet_{\mathrm{ch}}(A, A), A)$ is the boundary–bulk system.

Neither volume constructs a single chiral algebra. Affine Kac–Moody algebras arise from flat connections; Virasoro from reparametrizations; every other family in the standard landscape must be supplied by hand. The question is: what functor connects geometric input to chiral algebraic output, and what structure must it preserve?

The answer is the CY-to-chiral functor Φ. A Calabi–Yau category $\mathcal{C}$ carries a cyclic A_∞-structure encoding a d-dimensional Frobenius condition at the chain level. The functor converts this datum into a chiral algebra $A_{\mathcal{C}}$ on a curve, with three preservation properties: the bar complex of $A_{\mathcal{C}}$ recovers the cyclic bar complex of $\mathcal{C}$, the modular characteristic $\kappa_{\mathrm{ch}}(A_{\mathcal{C}})$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(\mathcal{C})$, and the $\mathbb{S}^d$-framing determines the operadic level.

1.2 The $\kappa_\bullet$-spectrum

A single CY manifold admits multiple chiral algebraizations. The lattice vertex algebra V_Λ attached to $H^*(K3,\mathbb{Z})$, the generalized Kac–Moody algebra whose denominator is the Igusa cusp form Δ_5, and the Conway module from the Leech lattice frame are three algebraizations of the same K3 surface. Each carries its own modular characteristic.

The $\kappa_\bullet$-spectrum records all four:

$$\begin{aligned}
\kappa_{\mathrm{cat}} &= \chi(\mathcal{O}_S) && \text{(holomorphic Euler characteristic)},\\
\kappa_{\mathrm{ch}} &= \dim_{\mathbb{C}} X && \text{(modular characteristic of the chiral de Rham complex)},\\
\kappa_{\mathrm{BKM}} &= \tfrac{1}{2}\,\mathrm{wt}(\Phi_{10}) && \text{(Borcherds–Kac–Moody automorphic weight)},\\
\kappa_{\mathrm{fiber}} &= \mathrm{rank}(\Lambda) && \text{(lattice rank)}.
\end{aligned}$$

For $K3\times E$: $\mathrm{Spec}_{\kappa_\bullet} = \{2,3,5,24\}$. The four numbers measure four faces of the same manifold: holomorphic Euler characteristic, boundary chiral algebra, bulk BKM superalgebra, and lattice frame. The chiral-holographic pair $(\kappa_{\mathrm{ch}}, \kappa_{\mathrm{BKM}}) = (3,5)$ governs the boundary–bulk axis of Volumes I–II. The $\kappa_\bullet$-spectrum is invisible from any single chiral algebra; it is intrinsic to the CY category.

1.3 The E_1 stabilization theorem

For $d \geq 3$, the CY chiral algebra is natively E_1, not E_d. The Schouten–Nijenhuis bracket on polyvector fields $\mathrm{PV}^*(\mathbb{C}^d)$ vanishes identically for $d \geq 3$: all noncommutative structure arises from quantization. The CoHA extension correspondence (sub before quotient) imposes a preferred ordering, which is E_1. The braided monoidal structure is recovered through the Drinfeld center of the E_1-monoidal representation category.

The obstruction to additional structure beyond E_1 is controlled by $\pi_d(BU)$, which is $\mathbb{Z}$ for even d and 0 for odd d by Bott periodicity. The effective obstruction is trivial precisely at $d \bmod 8 \in \{1,3,7\}$.

1.4 Outline

Section 2 reviews CY categories and cyclic A_∞-structures. Section 3 constructs the four-step functor Φ and proves Theorem CY-A$_2$. Section 4 defines the $\kappa_\bullet$-spectrum and develops its properties. Section 5 works through the $K3\times E$ prototype in detail: the thirteen structural results K3-1 through K3-13, the shadow–Siegel gap theorem, and the four obstructions separating κ_{ch} from κ_{BKM}. Sections 6–7 treat toric CY$_3$, the CoHA, and the E_1 stabilization theorem.

2 Calabi–Yau categories and cyclic A_∞-structures

2.1 CY categories

Definition 2.1 (Calabi–Yau category)**.** A smooth proper dg category $\mathcal{C}$ over a field k is *Calabi–Yau of dimension* d if there exists a quasi-isomorphism of $\mathcal{C}$-bimodules

$$\mathcal{C}^\vee \xrightarrow{\sim} \mathcal{C}[d]$$

where $\mathcal{C}^\vee = \mathrm{RHom}_{\mathcal{C}^e}(\mathcal{C}, \mathcal{C}^e)$ is the inverse dualizing bimodule and $[d]$ denotes the cohomological shift by d.

The CY condition encodes Serre duality at the categorical level. For the bounded derived category $D^b(\mathrm{Coh}(X))$ of a smooth projective variety X with trivial canonical bundle $\omega_X \simeq \mathcal{O}_X$, Serre duality gives $\mathbb{S}_X \simeq [d]$ on $D^b(\mathrm{Coh}(X))$ (where $d = \dim_{\mathbb{C}} X$), and the resulting CY_d-structure is the classical Serre functor.

Example 2.2 (Standard families)**.** Four families of CY categories provide the standard landscape:

(i) *Elliptic curves* ($d = 1$): $D^b(\mathrm{Coh}(E_\tau))$ is CY_1; the functor Φ produces the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$.

(ii) *K3 surfaces* ($d = 2$): $D^b(\mathrm{Coh}(S))$ is CY_2; Φ produces an E_2-chiral algebra with $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_S) = 2$.

(iii) *CY threefolds* ($d = 3$): the conifold, local $\mathbb{P}^2$, $K3 \times E$. The functor Φ at $d = 3$ is proved (Theorem CY-A$_3$).

(iv) *CY fourfolds* ($d = 4$): $K3 \times K3$, the sextic fourfold. The obstruction $\pi_4(BU) = \mathbb{Z}$ controls the shifted symplectic structure.

2.2 Cyclic A_∞-structure

A CY category $\mathcal{C}$ of dimension d carries a non-degenerate trace on its categorical Hochschild homology:

$$\mathrm{Tr}\colon \mathrm{HH}_d(\mathcal{C}) \longrightarrow k.$$

This trace encodes the CY condition as a d-dimensional Frobenius structure at the chain level. The primary invariant is the cyclic bar complex $\mathrm{CC}_\bullet(\mathcal{C})$, equipped with the Connes B-operator and an $\mathbb{S}^d$-framing.

Definition 2.3 (Cyclic A_∞-structure)**.** A *cyclic A_∞-structure* on a CY_d category $\mathcal{C}$ consists of:

(i) An A_∞-category structure on $\mathcal{C}$: operations $\mu_k\colon \mathcal{C}^{\otimes k} \to \mathcal{C}[2-k]$ for $k \geq 1$ satisfying the Stasheff identities.

(ii) A non-degenerate cyclic pairing $\langle -,- \rangle\colon \mathcal{C} \otimes \mathcal{C} \to k[-d]$ satisfying the higher cyclicity conditions $\langle \mu_k(a_1, \ldots, a_k), a_0 \rangle = \pm \langle \mu_k(a_0, a_1, \ldots, a_{k-1}), a_k \rangle$.

(iii) The induced Connes B-operator $B\colon \mathrm{CC}_n(\mathcal{C}) \to \mathrm{CC}_{n+1}(\mathcal{C})$ with $B^2 = 0$ and $Bb + bB = 0$.

Proposition 2.4 (Categorical Hochschild homology and the Gerstenhaber bracket)**.** *The categorical Hochschild cohomology $\mathrm{HH}^\bullet(\mathcal{C})$ of any dg category carries a Gerstenhaber bracket $[-,-]$ of degree -1. For a CY_d category, the CY pairing identifies $\mathrm{HH}^\bullet(\mathcal{C}) \simeq \mathrm{HH}_{d-\bullet}(\mathcal{C})$, and the Gerstenhaber bracket becomes a Lie bracket on $\mathrm{HH}_\bullet(\mathcal{C})$ of degree $d-1$.*

Proof. The Gerstenhaber bracket is a standard structure on categorical Hochschild cohomology (Gerstenhaber 1963, Keller 2006); the CY identification with homology uses the non-degenerate trace and the Hochschild–Kostant–Rosenberg degeneration at the E_1-page of the Hodge-to-de Rham spectral sequence. $\square$

2.3 The $\mathbb{S}^d$-framing

The CY_d-structure on $\mathcal{C}$ determines a framing of the categorical Hochschild complex by the d-sphere. The $\mathbb{S}^1$-action on $\mathrm{HH}_\bullet(\mathcal{C})$ (the Connes B-operator) is the $d = 1$ shadow; the full $\mathbb{S}^d$-action governs the operadic level of the chiral algebra.

Proposition 2.5 ($\mathbb{S}^d$-framing; Kontsevich–Vlassopoulos)**.** *For a* CY_d *category* $\mathcal{C}$*, the cyclic bar complex* $\mathrm{CC}_\bullet(\mathcal{C})$ *carries an* $\mathbb{S}^d$*-action compatible with the CY trace. At* $d = 2$*, this* $\mathbb{S}^2$*-action provides an* E_2*-algebra structure on the cyclic homology.*

The $\mathbb{S}^d$-framing is the datum that the functor Φ consumes to determine the operadic enhancement. At $d = 2$, the $\mathbb{S}^2$-framing is proved. At $d = 3$, the chain-level $\mathbb{S}^3$-framing compatible with the BV structure is proved by the ∞-categorical argument ($\mathrm{HH}^{-2}_{E_1} = 0$).

Remark 2.6 (Bott periodicity and the framing obstruction). The obstruction to extending the $\mathbb{S}^d$-framing beyond the base E_1 level is controlled by $\pi_d(BU)$:

d	Native E_n	$\pi_d(BU)$	Obstruction	Example
1	E_∞	0	trivial	elliptic curve
2	E_2	$\mathbb{Z}$	braiding (c_1)	K3, abelian surface
3	E_1	0	trivial	$K3 \times E$, conifold
4	E_1	$\mathbb{Z}$	shifted symplectic (p_1)	$K3 \times K3$

The pattern repeats with period 8 by Bott periodicity. For odd d, $\pi_d(BU) = 0$ by the parity of BU-homotopy groups ($\pi_k(BU) = \mathbb{Z}$ for even k, 0 for odd k; this follows from $\pi_k(BU) = \pi_{k-1}(U)$ and Bott periodicity for U).

3 The CY-to-chiral functor

3.1 The four-step construction

The functor $\Phi\colon \mathrm{CY}_d\text{-Cat} \to E_n\text{-ChirAlg}$ decomposes into four steps, each consuming a specific piece of the CY data and producing a specific algebraic structure.

Construction 3.1 (The functor Φ)**.** Let $\mathcal{C}$ be a smooth proper CY_d category and X a smooth curve.

Step 1. Cyclic A_∞ → Lie conformal algebra. The Gerstenhaber bracket on $\mathrm{HH}^\bullet(\mathcal{C})$ is a graded Lie bracket of degree -1 (Proposition 2.4). The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_\mathcal{C}$: the bracket becomes a λ-bracket $\{a_\lambda b\}$, and the pairing becomes the invariant bilinear form. This step consumes the A_∞-structure and the CY trace; it produces the current algebra of $\mathcal{C}$.

Step 2. Lie conformal algebra → factorization envelope. The factorization envelope construction produces a factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_\mathcal{C})$ on any smooth curve X. This step consumes the Lie conformal algebra and the curve; it produces a cosheaf on $\mathrm{Ran}(X)$ whose sections are the OPE data. The universal enveloping vertex algebra $U^{\mathrm{fact}}_X(\mathfrak{L}_\mathcal{C})$ is the chiral algebra.

Step 3. **$\mathbb{S}^d$-framing $\to$ E_n enhancement.** The $\mathbb{S}^d$-framing of $\mathrm{HH}_\bullet(\mathcal{C})$ (Proposition 2.5) enhances the factorization algebra to an E_n-chiral algebra. At $d = 2$: the $\mathbb{S}^2$-framing of Kontsevich–Vlassopoulos gives an E_2-chiral algebra. At $d \geq 3$: the native E_d-structure restricts to E_1 for CY quantum group purposes (the direct E_d-braiding is symmetric for $d \geq 3$, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$).

Step 4. **Quantization.** The CY trace $\mathrm{Tr}\colon \mathrm{HH}_d(\mathcal{C}) \to k$ determines a quantization of the classical factorization algebra. The quantum chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$ is the output. At $d = 2$, the Serre functor $\mathbb{S}_{\mathcal{C}} \simeq [2]$ kills the one-loop anomaly by parity, so the quantization preserves κ_{ch}. At $d = 3$, the Serre functor $\mathbb{S}_{\mathcal{C}} \simeq [3]$ no longer guarantees this; quantum corrections may shift κ_{ch}.

3.2 Theorem CY-A: the $d = 2$ case

Theorem 3.2 (CY-to-chiral functor at $d = 2$)**.** *There exists a functor $\Phi\colon \mathrm{CY}_2\text{-Cat} \to E_2\text{-ChirAlg}$ from smooth proper CY categories of dimension 2 to E_2-chiral algebras, such that:*

(i) $\Phi(\mathcal{C})$ is a chiral algebra whose generating fields are in bijection with $\mathrm{HH}^{\bullet+1}(\mathcal{C})$.

(ii) The bar complex $B(\Phi(\mathcal{C}))$ is quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(\mathcal{C})$ as a factorization coalgebra.

(iii) The modular characteristic satisfies $\kappa_{\mathrm{ch}}(\Phi(\mathcal{C})) = \chi^{\mathrm{CY}}(\mathcal{C}) := \sum_i (-1)^i \dim \mathrm{HH}_i(\mathcal{C})$.

Proof sketch. Steps 1–2 of Construction 3.1 are standard: the Gerstenhaber bracket produces a Lie conformal algebra, and the factorization envelope produces a chiral algebra. Step 3 at $d = 2$ uses the Kontsevich–Vlassopoulos $\mathbb{S}^2$-framing on $\mathrm{HH}_\bullet(\mathcal{C})$, which gives an E_2-algebra structure on the cyclic homology; this enhances the factorization algebra to an E_2-chiral algebra. Step 4 uses the CY trace to quantize; at $d = 2$, the Serre functor $\mathbb{S}_{\mathcal{C}} \simeq [2]$ forces the holomorphic anomaly to vanish (the one-loop correction from the quantization step involves $\mathbb{S}_{\mathcal{C}}[-d] = \mathbb{S}_{\mathcal{C}}[-2] \simeq \mathrm{Id}$, which is trivial). Part (i) follows from the construction; part (ii) from the identification of the bar differential with the cyclic boundary operator; part (iii) from the vanishing of the quantum correction. $\square$

Remark 3.3 (Verification at $d = 2$). For K3 surfaces: $\chi^{\mathrm{CY}}(D^b(\mathrm{Coh}(S))) = \chi(\mathcal{O}_S) = 2$, and $\kappa_{\mathrm{ch}} = 2$ is verified by six independent paths (direct Hodge, Mukai pairing, Dolbeault contracting homotopy, bar computation, Borcherds lift, elliptic genus). For abelian surfaces: $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_A) = 0$.

Conjecture 3.4 (CY modular characteristic at $d \geq 3$)**.** *For $\mathcal{C}$ a smooth proper CY$_d$ category with $d \geq 3$ (the functor Φ exists at $d = 3$ by Theorem CY-A$_3$; at $d \geq 4$ existence remains open),*

$$\kappa_{\mathrm{ch}}(\Phi(\mathcal{C})) \;=\; \chi^{\mathrm{CY}}(\mathcal{C}).$$

Remark 3.5 (Evidence). At $d = 3$, the conjecture is verified for $\mathbb{C}^3$ where both sides are independently computable ($\kappa_{\mathrm{ch}} = \kappa_{\mathrm{cat}} = 1$). The obstruction to extending the $d = 2$ proof is that the $d = 3$ quantization step may introduce corrections: Serre duality $\mathbb{S}_{\mathcal{C}} \simeq [3]$ no longer kills the one-loop anomaly by parity.

3.3 The five objects in the CY context

Under the functor Φ, the five-object chain of Volume I becomes a chain of CY invariants. The five objects are produced by four distinct functors and should never be conflated.

Remark 3.6 (Five objects from $\Phi(\mathcal{C})$). Let $A_{\mathcal{C}} = \Phi(\mathcal{C})$ be the chiral algebra of a CY$_d$ category $\mathcal{C}$.

(i) $A_{\mathcal{C}}$ (the chiral algebra): the factorization envelope of the Lie conformal algebra $\mathfrak{L}_{\mathcal{C}}$.

(ii) $B(A_{\mathcal{C}}) = T^c(s^{-1}\bar{A}_{\mathcal{C}})$ (the bar coalgebra): a factorization coalgebra on $\mathrm{Ran}(X)$ with deconcatenation coproduct. At genus $g \geq 1$, the fiberwise differential satisfies $d_{\mathrm{fib}}^2 = \kappa_{\mathrm{ch}}(A_{\mathcal{C}}) \cdot \omega_g$.

(iii) $A_{\mathcal{C}}^{i} = H^*(B(A_{\mathcal{C}}))$ (the dual coalgebra): its coalgebra structure encodes the BPS spectrum of $\mathcal{C}$.

(iv) $A_{\mathcal{C}}^! = ((A_{\mathcal{C}}^{i})^{\vee})$ (the Koszul dual algebra): obtained by Verdier duality D_{Ran}. For CY$_2$ categories, $A_{\mathcal{C}}^!$ is the chiral algebra of the mirror CY$_2$.

(v) $Z_{\mathrm{ch}}^{\mathrm{der}}(A_{\mathcal{C}})$ (the derived chiral center): the chiral Hochschild cochain complex encoding the bulk sector.

The bar complex is the E_1 engine; the derived center is the $\mathrm{SC}^{\mathrm{ch,top}}/E_3$ output. Bar-cobar $\Omega(B(A_{\mathcal{C}})) \xrightarrow{\sim} A_{\mathcal{C}}$ is inversion (recovering the algebra), not Koszul duality; the derived center is the categorical Hochschild construction, not the output of bar-cobar.

4 The $\kappa_\bullet$-spectrum

4.1 Four modular characteristics

A CY manifold X admits multiple chiral algebraizations, each with its own modular characteristic. The four approved subscripts in this volume are: κ_{ch} (chiral), κ_{cat} (categorical), κ_{BKM} (Borcherds–Kac–Moody), and κ_{fiber} (lattice). Bare κ without subscript is forbidden.

Definition 4.1 (The $\kappa_\bullet$-spectrum)**.** For a CY$_d$ manifold X (or more generally a CY$_d$ category $\mathcal{C}$), the *$\kappa_\bullet$-spectrum* is the ordered tuple

$$\mathrm{Spec}_{\kappa_\bullet}(X) = (\kappa_{\mathrm{cat}},\ \kappa_{\mathrm{ch}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}})$$

where:

(i) $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_X)$ (the holomorphic Euler characteristic) measures the categorical categorical Hochschild data. This is the multiplier in the BPS degeneracy factorization $A_n = \kappa_{\mathrm{cat}} \cdot \dim(\rho_n)$.

(ii) $\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}(A_{\mathcal{C}})$ (the modular characteristic of the boundary chiral algebra) governs genus-1 curvature: $F_1 = \kappa_{\mathrm{ch}}/24$. For $d = 2$, $\kappa_{\mathrm{ch}} = \chi^{\mathrm{CY}}(\mathcal{C}) = \chi(\mathcal{O}_X)$ (Theorem 3.2).

(iii) $\kappa_{\mathrm{BKM}} = \frac{1}{2}\,\mathrm{wt}(\Phi_{\mathrm{aut}})$ (half the weight of the automorphic denominator form) measures the bulk Borcherds–Kac–Moody superalgebra. For $K3 \times E$: $\kappa_{\mathrm{BKM}} = \frac{1}{2}\,\mathrm{wt}(\Phi_{10}) = 5$.

(iv) $\kappa_{\mathrm{fiber}} = \mathrm{rank}(\Lambda_X)$ (the lattice rank) measures the frame data. For $K3 \times E$: $\kappa_{\mathrm{fiber}} = 24$.

The four characteristics are generically distinct. Their relations encode the internal structure of the CY manifold.

Proposition 4.2 (Spectrum relations for $K3 \times E$)**.** *For $X = K3 \times E$:*

$$\mathrm{Spec}_{\kappa_\bullet}(K3 \times E) = (2,\ 3,\ 5,\ 24).$$

The first three satisfy the second-quantization relation

$$\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{cat}} + \kappa_{\mathrm{ch}}$$

and the lattice relation $\kappa_{\mathrm{fiber}} = 12 \cdot \kappa_{\mathrm{cat}}$. The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects the single-copy-to-second-quantization passage.

Proof. $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3}) = 2$ (Noether's formula). $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = \dim_{\mathbb{C}}(K3 \times E) = 3$ (additivity under Cartesian product, Route B: $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3) = 2$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(E) = 1$; verified by six independent paths). This is the Heisenberg-level additive invariant; the Hodge-supertrace Route A $\Xi = \kappa_{\mathrm{ch}}$ is Künneth-multiplicative and vanishes on $K3 \times E$. See Volume III Remark **??** for the adjudication. $\kappa_{\mathrm{BKM}} = \frac{1}{2}\,\mathrm{wt}(\Phi_{10}) = 5$ ($\Phi_{10} = \Delta_5^2$; the Igusa cusp form has weight 10, so $\kappa_{\mathrm{BKM}} = 5$). $\kappa_{\mathrm{fiber}} = \mathrm{rank}(H^*(K3, \mathbb{Z})) = 24$ (the second Betti number $b_2(K3) = 22$ plus $b_0 + b_4 = 2$). The second-quantization relation $5 = 2 + 3$ is verified. The lattice relation $24 = 12 \cdot 2$ uses $\chi(K3) = 24 = 12 \cdot \kappa_{\mathrm{cat}}$ (Noether for K3). $\square$

4.2 Koszul complementarity in the CY context

Volume I defines the Koszul conductor $K(A) = \kappa_{\mathrm{ch}}(A) + \kappa_{\mathrm{ch}}(A^!)$: it is 0 for KM/Heisenberg/lattice/free families, 13 for Virasoro, and family-dependent in general. The CY context introduces a new complementarity: mirror symmetry exchanges $\mathcal{C}$ and $\mathcal{C}^\vee$, and $A_{\mathcal{C}^\vee} = A^!_{\mathcal{C}}$ when both are constructed.

Conjecture 4.3 (CY Koszul complementarity)**.** *For a CY_2 category $\mathcal{C}$ with mirror $\mathcal{C}^\vee$:*

$$\kappa_{\mathrm{ch}}(\Phi(\mathcal{C})) + \kappa_{\mathrm{ch}}(\Phi(\mathcal{C}^\vee)) = 0.$$

The Koszul conductor vanishes for all mirror pairs at $d = 2$.

Remark 4.4 (Evidence). For the quintic $X_5 \subset \mathbb{P}^4$: $\kappa_{\mathrm{cat}}(X_5) = \chi(X_5)/2 = -100$ and $\kappa_{\mathrm{cat}}(X_5^\vee) = +100$; the sum $\kappa_{\mathrm{cat}}(X_5) + \kappa_{\mathrm{cat}}(X_5^\vee) = 0$ is consistent with vanishing Koszul conductor for the KM/free family pattern of Volume I.

4.3 Shadow depth classification

The CY-to-chiral functor assigns to each CY geometry a chiral algebra whose shadow class (Volume I) organizes the landscape.

Proposition 4.5 (Shadow depth of CY chiral algebras)**.** *Under Φ, the shadow depth classification of Volume I (G/L/C/M) distributes over the CY landscape as follows:*

CY geometry	*Shadow class*	κ_{ch}	*Shadow depth*
$\mathbb{C}^3$ *(toric, abelian)*	G	1	2
Resolved conifold	G	1	2
Local $\mathbb{P}^2$	M	3/2	∞
$K3 \times E$	M	3	∞

Classes G and M are populated. Classes L (shadow depth 3*) and C (shadow depth* 4*) are targets of the toric programme: the toric vertex bound $r_{\max} \leq N$ (where N is the number of web-diagram vertices) predicts that specific toric geometries should realize these classes, but explicit CY examples at shadow depth exactly* 3 *or* 4 *remain to be identified.*

5 The $K3 \times E$ prototype

The product of a K3 surface S and an elliptic curve E is the canonical CY_3 testing ground. It is simple enough to compute (lattice $\Lambda^{3,2}$ of signature $(3, 2)$, K3 elliptic genus $\phi_{0,1}(\tau, z)$, Borcherds denominator Δ_5) and already exhibits every obstruction that separates the shadow tower from the full automorphic form.

5.1 The thirteen structural results

We record the first six of the thirteen structural results; the remaining seven concern mock modularity, scattering diagrams, and the Schottky programme and will appear in the full version.

K3-1 (Heisenberg-level modular characteristic, Route B): $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by six independent paths (direct Hodge, Mukai pairing, Dolbeault contracting homotopy, bar computation, Borcherds lift, elliptic genus). Additivity under Cartesian product: $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3) = 2$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(E) = 1$. The Route A Hodge-supertrace $\Xi = \kappa_{\mathrm{ch}}$ is Künneth-multiplicative and gives $\Xi(K3 \times E) = 2 \cdot 0 = 0$; the two routes label genuinely distinct invariants. See Volume III Remark **??** for the full adjudication.

K3-2 (moonshine multiplier): the factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is $\kappa_{\mathrm{cat}}(K3) = \chi(\mathcal{O}_{K3}) = 2$. The massive Mathieu multiplicities satisfy $A_n = \kappa_{\mathrm{cat}} \cdot \dim(\rho_n)$: the bar complex provides the universal coefficient and does not detect M_{24} directly.

K3-3 (BKM weight): $\kappa_{\mathrm{BKM}} = \frac{1}{2}\,\mathrm{wt}(\Phi_{10}) = 5$. The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects second quantization; the Hilbert–Chow exceptional divisor accounts for $\chi(\mathrm{Hilb}^2) - \chi(\mathrm{Sym}^2) = 24 = \chi(K3)$.

K3-4 (BPS factorization): $A_n = \kappa_{\mathrm{cat}} \cdot \dim(\rho_n)$, separating the homological part ($\kappa_{\mathrm{cat}} = 2$) from the group-theoretic part (ρ_n, the Mathieu representation).

K3-5 (mock modular verification): the mock modular decomposition $Z_{K3} = (h - 24\mu)\,\vartheta_1^2/\eta^3$ is verified to machine precision ($\sim 5.7 \times 10^{-16}$ relative error), where μ is the Zwegers Appell–Lerch sum.

K3-6 (shadow convergence): the shadow generating function $Z^{\mathrm{sh}}(t) = \kappa_{\mathrm{ch}}\big((t/2)/\sin(t/2) - 1\big)$ converges with radius $R = 2\pi$, has entire Borel transform, and exhibits no resurgence.

5.2 The shadow–Siegel gap theorem

Theorem 5.1 (Shadow–Siegel gap)**.** *The shadow obstruction tower of $K3\times E$ does not produce the Igusa cusp form Φ_{10}. Four structural obstructions separate the two:*

(O1) Categorical*: the shadow tower produces numbers (Taylor coefficients of Θ_{A_C}); the denominator identity is a function (an automorphic form on $\mathcal{H}_2$).*

(O2) κ_{ch}–κ_{BKM} mismatch*: $\kappa_{\mathrm{ch}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$; the single-copy chiral algebra sees $\dim_{\mathbb{C}} X$, while the bulk reconstruction sees the Borcherds weight.*

(O3) Second quantization*: the physical DT partition function is the DMVV symmetric product of single-copy data; the bar complex of a single algebra is the single copy.*

(O4) Schottky*: at $g \geq 4$, the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ is no longer dominant, and the shadow tower is imprisoned in tautological classes on $\overline{\mathcal{M}}_g$.*

Each obstruction points to a research programme. The κ_{ch}–κ_{BKM} mismatch (O2) is the most telling: a single CY_3 admits multiple chiral algebraizations, and the $\kappa_\bullet$-spectrum records which face of the geometry each algebraization captures.

6 Toric CY_3 and the cohomological Hall algebra

A toric CY_3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram fixes a quiver with potential (Q_X, W_X), and the critical CoHA $\mathcal{H}(Q_X, W_X) = \bigoplus_{\mathbf{d}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$ provides the E_1-sector of the CY quantum group.

6.1 $\mathbb{C}^3$ and the affine Yangian

The simplest toric CY_3. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop $\circlearrowleft$). Cubic potential $W = \mathrm{tr}(XYZ - XZY)$.

Theorem 6.1 (Schiffmann–Vasserot)**.** *The critical CoHA of $\mathbb{C}^3$ is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:*

$$\mathrm{CoHA}(\mathbb{C}^3) \;\simeq\; Y^+(\widehat{\mathfrak{gl}}_1) \;\simeq\; \mathcal{W}_{1+\infty}.$$

The chain $\mathbb{C}^3 \to \mathcal{W}_{1+\infty} \to \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ is verified end-to-end: the toric CY_3 produces the CoHA, which is the positive half of the Yangian, which is the E_1-chiral quantum group. The Drinfeld center recovers the full E_2-braiding (Theorem 6.3 below).

6.2 The CoHA as E_1-primitive

The CoHA is an associative (E_1) algebra: the Hall multiplication is ordered (short exact sequences $0 \to V' \to V \to V'' \to 0$ have sub before quotient). In the present framework:

- The CoHA = the positive half of $G(X)$ = the E_1-chiral sector (the ordered part).
- The full quantum vertex chiral group $G(X)$ is E_2 (braided).

- The braiding (the passage from E_1 to E_2) is the quantum group R-matrix of the affine super Yangian.
- The ordered bar $B^{\mathrm{ord}}(A_{\mathcal{C}})$ preserves the R-matrix; the symmetric bar $B^{\Sigma}(A_{\mathcal{C}})$ kills the Hopf structure via averaging: in abelian and scalar families $\mathrm{av}(r(z)) = \kappa_{\mathrm{ch}}$, while for non-abelian affine Kac–Moody $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa_{\mathrm{ch}}$.

Remark 6.2 (Shadow = $\mathrm{GW}(\mathbb{C}^3)$). At $\kappa_{\mathrm{ch}} = \Psi$ (the deformation parameter of $\mathcal{W}_{1+\infty}$ at $c = 1$), the shadow obstruction tower of Volume I produces the perturbative constant-map Gromov–Witten free energies $F_g^{\mathrm{GW,const}}(\mathbb{C}^3)$: the shadow is the full GW answer for $\mathbb{C}^3$ (which has no compact curves, hence no worldsheet instantons). On the DT side, the MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ encodes the partition function via the MNOP correspondence.

6.3 The Drinfeld center

Theorem 6.3 (Drinfeld center for $\mathbb{C}^3$)**.** *The Drinfeld center of the E_1-monoidal representation category of $Y^+(\widehat{\mathfrak{gl}}_1)$ recovers the full E_2-braided monoidal structure:*

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \;\simeq\; \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \;\simeq\; \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

Five invariants match at six levels: the naive center has dimension 1*, while the derived center has dimension* 3 *(capturing* $\mathrm{Ext}^{1,2}$ *invisible to the commutant).*

The Drinfeld center is the categorified averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ of Volume I: it projects the ordered R-matrix data to the scalar κ_{ch} at the level of the $\kappa_\bullet$-spectrum, while recovering non-trivial braiding at the categorical level.

7 The E_1 stabilization theorem

Theorem 7.1 (E_1 stabilization for $d \geq 3$)**.** *For $d \geq 3$, the CY chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$ is natively E_1. The reason is structural: the* $\mathrm{GL}(d)$*-invariant Schouten–Nijenhuis bracket on polyvector fields* $\mathrm{PV}^*(\mathbb{C}^d)$ *vanishes identically for $d \geq 3$, so the classical Lie conformal algebra is abelian and all noncommutative structure arises from quantization. The CoHA extension correspondence imposes a preferred ordering (sub before quotient), which is E_1, not E_2. The braided monoidal structure is recovered through the Drinfeld center.*

Proof sketch. The Schouten–Nijenhuis bracket $[-,-]_{\mathrm{SN}}\colon \mathrm{PV}^p(\mathbb{C}^d)\otimes\mathrm{PV}^q(\mathbb{C}^d) \to \mathrm{PV}^{p+q-1}(\mathbb{C}^d)$ requires $p+q-1 \leq d$ for non-trivial output (the bracket of two polyvectors of total degree exceeding $d+1$ vanishes by degree reasons on a d-dimensional manifold). At $d = 2$: $\mathrm{PV}^2(\mathbb{C}^2) = \mathbb{C}$ is one-dimensional and the bracket $[f\partial_1 \wedge \partial_2, g\partial_1 \wedge \partial_2] = 0$ is trivially zero; the non-trivial bracket lives at $(p,q) = (1,1)$, producing the Poisson bracket. At $d \geq 3$: the antisymmetry of the Schouten–Nijenhuis bracket combines with the $\mathrm{GL}(d)$-invariance condition to force the bracket to vanish on the CY locus. The quantization then proceeds through the CoHA, which is associative (E_1): the extension correspondence imposes a preferred ordering, and no additional Lie-theoretic data survives. □

Remark 7.2 (Arithmetic of the fundamental group). The key arithmetic: $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) = \mathbb{Z}$ (braid group, native E_2, non-symmetric), while $\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$ for $d \geq 3$ (symmetric group). Dimension $d = 2$ is the sweet spot where the native E_2-structure already carries a non-symmetric braiding. For $d \geq 3$, the descent-and-center detour is mandatory: one must discard the over-symmetric E_d-structure, pass to E_1, and rebuild the braiding categorically through the Drinfeld center.

8 Wave-14 reconception: Φ scope, CY-D stratification, universal trace

The CY-to-chiral functor Φ is sharpened against three 2026-04-17 platonic-ideal anchors: the explicit E_n-output scope (*FM43*), the dimension-stratified κ_{ch} spectrum, and the universal trace identity connecting the Vol I conductor $K = -c_{\mathrm{ghost}}$ with the Vol III Borcherds $\kappa_{\mathrm{BKM}} = c_N(0)/2$.

Theorem 8.1 (E_n output scope of Φ; FM43)**.** *The CY-to-chiral functor* $\Phi\colon \mathrm{CY}_d\text{-Cat} \longrightarrow E_n\text{-ChirAlg}$ *has*

$$n \;=\; \begin{cases} 2 & d \leq 2, \\ 1 & d \geq 3. \end{cases}$$

At $d \leq 2$ the native $\mathbb{S}^2$-framing on $\mathrm{HH}_\bullet(\mathcal{C})$ *supplies a genuine E_2-chiral structure (Kontsevich–Vlassopoulos; Theorem* CY-A$_2$*). At $d \geq 3$ the E_d-structure becomes symmetric ($\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$), and the functor output is natively E_1-chiral; the braided monoidal structure is recovered* categorically *via the Drinfeld centre, closing the E_n operadic circle on $Z^{\mathrm{der}}_{\mathrm{ch}}$ rather than on $\Phi(\mathcal{C})$.*

Theorem 8.2 (κ_{ch} stratification by CY-dimension)**.** *For a compact Calabi–Yau manifold X of dimension d,*

$$\kappa_{\mathrm{ch}}(\Phi(D^b\mathrm{Coh}(X))) \;=\; \sum_{q=0}^{d} (-1)^q h^{0,q}(X), \tag{1}$$

the Hodge supertrace. The stratification across the canonical families reads

family (dim d)	κ_{ch}	*notes*
elliptic curve E ($d=1$)	0	$h^{0,0} - h^{0,1} = 1 - 1$
K3 ($d=2$)	2	$\chi(\mathcal{O}_{K3}) = 2$
abelian/bielliptic surface ($d=2$)	0	$\chi(\mathcal{O}) = 0$
quintic $\subset \mathbb{P}^4$ ($d=3$)	0	$\chi(\mathcal{O}_X) = 0$
$K3 \times E$ ($d=3$)	0	$\chi(\mathcal{O}) = 0$
triple elliptic E^3 ($d=3$)	0	$\chi(\mathcal{O}) = 0$
local $\mathbb{P}^2 = \mathrm{Tot}(K_{\mathbb{P}^2})$	3/2	$\chi(S)/2 = 3/2$ *(shadow)*
CY_4 *sextic $\subset \mathbb{P}^5$*	2	*Hodge supertrace*
generic CY_5	0	*Hodge supertrace vanishes*

The conifold is not *a local surface: its geometry is* $\mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1)$*, not* $\mathrm{Tot}(K_S \to S)$*, and the identity $\kappa_{\mathrm{ch}} = \chi(S)/2$ does not apply. The McKay orbifold $[\mathbb{C}^3/\mathbb{Z}_3]$ gives $\kappa_{\mathrm{ch}} = 1$ by direct calculation.*

Proof sketch. The identification $\kappa_{\mathrm{ch}} = \sum(-1)^q h^{0,q}$ is the Hochschild–Kostant–Rosenberg degeneration composed with the Mukai pairing; the degree-d cyclic trace $\mathrm{Tr}\colon \mathrm{HC}^-_d(\mathcal{C}) \to k$ restricts to the top-dimensional Hodge piece, and Serre duality pairs it with the full Hodge decomposition. The family entries are direct computation; "local $\mathbb{P}^2$" comes from the shadow $\kappa_{\mathrm{ch}} = \chi(S)/2 = 3/2$ because $\chi(\mathbb{P}^2) = 3$. The conifold exclusion is structural: its total space is not the total space of a canonical line bundle over a surface, so the local-surface formula is inapplicable. □

Theorem 8.3 (Universal trace identity: $K = -c_{\mathrm{ghost}} = 2\,\kappa_{\mathrm{BKM}}$)**.** *On the overlap of the Vol I standard landscape and the Vol III Borcherds-admissible locus (the Fricke-type paramodular locus $N \in \{1, 2, 3, 4, 6\}$), the Koszul conductor, the BRST ghost charge, and the Borcherds weight unify in a single identity:*

$$K(\mathcal{A}) \;=\; -\,c_{\mathrm{ghost}}(\mathcal{A}) \;=\; 2\,\kappa_{\mathrm{BKM}}(\Phi_N), \tag{2}$$

where $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the Φ_N-leading Fourier coefficient of the Borcherds product defining $\mathcal{A}^!$.

Proof sketch. The Vol I identity $K = -c_{\mathrm{ghost}}$ is Theorem platonic-conductor (see companion `chiral_chern_weil.tex` addendum). The Vol III identity $\kappa_{\mathrm{BKM}} = c_N(0)/2$ is Theorem `borcherds-weight-kappa-BKM-universal` of `cy_d_kappa_stratification.tex` (Vol III). Matching the two through the bridge $c_{\mathrm{ghost}} = -2\kappa_{\mathrm{BKM}}$ is a numerical verification across the five paramodular levels; at $N = 1$ one recovers Monstrous moonshine ($\kappa_{\mathrm{BKM}}(\Phi_1) = 12 = -c_{\mathrm{ghost}}/2$ for the Monster module; complementarity $K = 24$ on the Leech shadow). □

Remark 8.4 (κ subscript conventions (all subscripts explicit)). Every κ in this addendum carries an approved Vol III subscript from the closed set $\{\mathrm{ch}, \mathrm{cat}, \mathrm{BKM}, \mathrm{fiber}\}$. Bare κ is forbidden in Vol III. The identifications $\kappa_{\mathrm{ch}} \to \chi(\mathcal{O})$ (Hodge supertrace), $\kappa_{\mathrm{BKM}} \to c_N(0)/2$, and the bridge $K = -c_{\mathrm{ghost}} = 2\kappa_{\mathrm{BKM}}$ are the three precise uses in this section.

Remark 8.5 (Epistemic tags: Φ status). Φ at $d = 1$: trivial ($A_\infty = \mathrm{comm}$). $d = 2$: *proved*, Theorem CY-A$_2$, E_2-chiral output. $d = 3$: *proved*, Theorem CY-A$_3$, ∞-categorical, E_1-chiral output. $d = 4$: obstructed by $\pi_4(BU) = \mathbb{Z}$; the p_1-twisted double current algebra is proved where the Pontryagin obstruction vanishes (e.g. $K3 \times K3$); elsewhere conjectural. $d = 5, 6$: open; conjectural Hodge supertrace formula recovers $\kappa_{\mathrm{ch}} = 0$ for generic smooth CY. The unconditional range is $d \in \{1, 2, 3\}$; $d = 4$ is family-conditional.

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